Mack's chain-ladder model in Lean 4: blueprint

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Contents

Chapter 1

Introduction

This blueprint tracks the formalization of Mack's distribution-free chain-ladder model [1] in Lean 4 on top of mathlib. Each statement below carries the name of its Lean declaration; green nodes in the dependency graph are proved, the others are open work listed in the roadmap. Statements are given in the paper's own terms, and where a published formula is an estimator or an approximation rather than an identity, the text says so.

A run-off triangle with n accident years and n development years is a function $C : \mathbb{N} \to \mathbb{N} \to \mathbb{R}$; $C_{i,k}$ is the cumulative claims of accident year i at development year k, observed when $i+k \leq n-1$. The latest observed entry of accident year i is $C_{i,\,n-1-i}$.

Chapter 2

The chain-ladder estimators

2.1 Definitions

Definition 2.1 (Contributors)**.** The accident years contributing to the development factor $k \to k+1$ are $i = 0, \dots, n-k-2$, those with $C_{i,k+1}$ observed.

Definition 2.2 (Column sums)**.** $S_k = \sum_{i=0}^{n-k-2} C_{i,k}$ and $T_k = \sum_{i=0}^{n-k-2} C_{i,k+1}$.

Definition 2.3 (Development factor)**.** $\hat{f}_k = T_k / S_k$. (In Lean, division by zero is zero.)

Definition 2.4 (Individual factor)**.** $F_{i,k} = C_{i,k+1}/C_{i,k}$.

Definition 2.5 (Mack's variance estimator)**.** $\hat{\sigma}_k^2 = \frac{1}{n-k-2} \sum_{i=0}^{n-k-2} C_{i,k} \, (F_{i,k} - \hat{f}_k)^2$. The value for $k = n-2$ (a single contributor) is an extrapolation convention in [1], not an estimator; no theorem here uses it.

Definition 2.6 (Projection)**.** $\hat{C}_{i,k} = C_{i,n-1-i} \prod_{j=n-1-i}^{k-1} \hat{f}_j$ for $k \geq n-1-i$.

Definition 2.7 (Ultimate and reserve)**.** $\hat{C}_{i,n-1}$ is the chain-ladder ultimate and $\hat{R}_i = \hat{C}_{i,n-1} - C_{i,n-1-i}$ the reserve.

Definition 2.8 (Mack's MSEP estimator)**.** Mack (1993), Theorem 3:

$$\widehat{\mathrm{msep}}(\hat{R}_i) = \hat{C}_{i,n-1}^2 \sum_{k=n-1-i}^{n-2} \frac{\hat{\sigma}_k^2}{\hat{f}_k^2} \Big(\frac{1}{\hat{C}_{i,k}} + \frac{1}{S_k} \Big).$$

2.2 Deterministic identities

Lemma 2.9 (Weighted-average form)**.** *If $C_{i,k} \neq 0$ for every contributor i, then $\hat{f}_k = \sum_i \frac{C_{i,k}}{S_k} F_{i,k}$.*

Proof. Termwise: $\frac{C_{i,k}}{S_k} \cdot \frac{C_{i,k+1}}{C_{i,k}} = \frac{C_{i,k+1}}{S_k}$. □

Lemma 2.10. *If $C_{i,k} \neq 0$ for every contributor, $T_k = \sum_i C_{i,k} F_{i,k}$.*

Proof. Cancel the denominator termwise. □

Lemma 2.11 (Sum-of-squares decomposition)**.** *For any centre f, with $C_{i,k} \neq 0$ for every contributor, $\sum_i C_{i,k} (F_{i,k} - f)^2 = \sum_i C_{i,k} F_{i,k}^2 - 2 f T_k + f^2 S_k$.*

Proof. Expand the square and use Lemma 2.10. $\square$

Lemma 2.12 (Collapse at the chain-ladder centre). *If moreover $S_k \neq 0$, then $\sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2 = \sum_i C_{i,k}F_{i,k}^2 - S_k\hat{f}_k^2$. This is the identity behind the $n-k-2$ degrees of freedom in $\hat{\sigma}_k^2$.*

Proof. Put $f = \hat{f}_k = T_k/S_k$ in Lemma 2.11. $\square$

Lemma 2.13 (Projection step). *For $k \geq n-1-i$, $\hat{C}_{i,k+1} = \hat{C}_{i,k}\,\hat{f}_k$.*

Proof. Split off the top factor of the product. $\square$

Lemma 2.14. $\hat{C}_{i,n-1-i} = C_{i,n-1-i}$.

Proof. Empty product. $\square$

Lemma 2.15. *The oldest accident year is fully developed: $\hat{R}_0 = 0$.*

Proof. Empty product again. $\square$

Chapter 3

Mack 1999 equals Mack 1993

Definition 3.1 (Mack's 1999 recursion)**.** For $\alpha = 1$ and unit weights [2], starting from $\mathrm{se}^2 = 0$ on the latest diagonal $d = n - 1 - i$ and stepping m times,

$$\mathrm{se}^2_{m+1} = \hat{C}^2_{i,d+m}\Big(\frac{\hat{\sigma}^2_{d+m}}{\hat{C}_{i,d+m}} + \frac{\hat{\sigma}^2_{d+m}}{S_{d+m}}\Big) + \mathrm{se}^2_m\,\hat{f}^2_{d+m}.$$

Definition 3.2. The summand of the closed form: $\mathrm{term}_{i,k} = \frac{\hat{\sigma}^2_k}{\hat{f}^2_k}\big(\frac{1}{\hat{C}_{i,k}} + \frac{1}{S_k}\big)$.

Lemma 3.3. $\widehat{\mathrm{msep}}(\hat{R}_i) = \hat{C}^2_{i,n-1}\sum_{k=n-1-i}^{n-2}\mathrm{term}_{i,k}$.

Proof. By definition. □

Lemma 3.4 (Truncated closed form)**.** *If $\hat{f}_k \neq 0$ for $d \le k < d{+}m$, then* $\mathrm{se}^2_m = \hat{C}^2_{i,d+m}\sum_{k=d}^{d+m-1}\mathrm{term}_{i,k}$.

Proof. Induction on m using Lemma 2.13 and $\hat{f}^2_{d+m}/\hat{f}^2_{d+m} = 1$. □

Theorem 3.5 (Mack 1999 = Mack 1993)**.** *For $i \le n-1$ and $\hat{f}_k \neq 0$ along the row, i steps of the 1999 recursion return exactly Mack's 1993 estimator:* $\mathrm{se}^2_i = \widehat{\mathrm{msep}}(\hat{R}_i)$.

Proof. Lemma 3.4 with $m = i$, since $d + i = n - 1$. □

Chapter 4

Bornhuetter-Ferguson on the chain-ladder pattern

Definition 4.1 (Cumulative development factor, BF reserve and ultimate)**.** $\mathrm{cdf}_i = \prod_{k=d}^{n-2} \hat{f}_k$; for an a priori ultimate U, $R_i^{BF} = U\,(1 - 1/\mathrm{cdf}_i)$ and $\hat{C}_{i,n-1}^{BF} = C_{i,d} + R_i^{BF}$.

Lemma 4.2 (Linearity)**.** *$R_i^{BF}(aU) = a\,R_i^{BF}(U)$ and $R_i^{BF}(U+V) = R_i^{BF}(U) + R_i^{BF}(V)$.*

Proof. Ring identities. □

Theorem 4.3 (BF with the chain-ladder prior is chain ladder)**.** *If $\mathrm{cdf}_i \neq 0$ and $U = \hat{C}_{i,n-1}$, then $\hat{C}_{i,n-1}^{BF} = \hat{C}_{i,n-1}$ and $R_i^{BF} = \hat{R}_i$.*

Proof. $\hat{C}_{i,n-1} = C_{i,d}\,\mathrm{cdf}_i$, so $U(1 - 1/\mathrm{cdf}_i) = C_{i,d}\mathrm{cdf}_i - C_{i,d}$. □

Lemma 4.4 (The BF reserve is a fraction of the prior)**.** *If every development factor along the row is at least 1, then $\mathrm{cdf}_i \geq 1$ and, for $U \geq 0$, $0 \leq R_i^{BF}(U) \leq U$.*

Proof. A product of numbers at least 1 is at least 1; then $0 \leq 1 - 1/\mathrm{cdf}_i \leq 1$. □

Chapter 5

The stochastic layer

5.1 Model

Definition 5.1 (Random triangle)**.** A random run-off triangle on a measurable space Ω is a family of random variables $C_{i,k} : \Omega \to \mathbb{R}$ together with a filtration $\mathcal{D}_k$ such that $C_{i,j}$ is $\mathcal{D}_k$-measurable whenever $j \le k$ ("everything observed up to development year k").

Definition 5.2 (Assumption (M1))**.** $E[C_{i,k+1} \mid \mathcal{D}_k] = f_k\, C_{i,k}$ almost surely, for all i and k. This is Mack's first assumption in the form conditioned on $\mathcal{D}_k$; Mack states it conditioned on the row's own history and obtains this form from independence across accident years (M2), a passage still to be formalized.

Definition 5.3 (Estimators as random variables)**.** $S_k(\omega)$ and $\hat{f}_k(\omega)$ are the deterministic estimators applied to the triangle observed at ω.

Lemma 5.4. *S_k is $\mathcal{D}_k$-measurable.*

Proof. A finite sum of $\mathcal{D}_k$-measurable functions. □

5.2 Mack's Theorem 2

Theorem 5.5 (Conditional unbiasedness of the development factor)**.** *Assume (M1), that $S_k \neq 0$ almost surely, that each $C_{i,k+1}$ and $\hat{f}_k$ are integrable. Then $E[\hat{f}_k \mid \mathcal{D}_k] = f_k$ almost surely.*

Proof. $\hat{f}_k = S_k^{-1} \sum_i C_{i,k+1}$. The factor S_k^{-1} is $\mathcal{D}_k$-measurable and comes out of the conditional expectation; by (M1) the conditional expectation of the sum is $f_k \sum_i C_{i,k} = f_k S_k$; cancel S_k where it is nonzero. □

Lemma 5.6. *$\hat{f}_j$ is $\mathcal{D}_k$-measurable whenever $j + 1 \le k$.*

Proof. S_j is $\mathcal{D}_j \subseteq \mathcal{D}_k$-measurable and each $C_{i,j+1}$ is $\mathcal{D}_k$-measurable. □

Theorem 5.7 (Uncorrelated development factors, conditional form)**.** *Under (M1), on a finite measure space, for $j < k$ (with the integrability and $S \neq 0$ hypotheses of Theorem 5.5 for both columns and $\hat{f}_j \hat{f}_k$ integrable), $E[\hat{f}_j \hat{f}_k \mid \mathcal{D}_j] = f_j f_k$ almost surely.*

Proof. Condition on $\mathcal{D}_k$ first: $\hat{f}_j$ is $\mathcal{D}_k$-measurable and comes out, and $E[\hat{f}_k \mid \mathcal{D}_k] = f_k$ by Theorem 5.5. Then the tower property down to $\mathcal{D}_j$ and Theorem 5.5 again for column j. □

Theorem 5.8 (Mack 1993, Theorem 2). *On a probability space, under the same hypotheses, $E[\hat{f}_k] = f_k$ and $E[\hat{f}_j \hat{f}_k] = f_j f_k$ for $j < k$: the chain-ladder factors are unbiased and uncorrelated, which is what makes $\prod_k \hat{f}_k$ an unbiased estimator of $\prod_k f_k$.*

Proof. Integrate the conditional statements. □

5.3 Mack's Theorem 1

Definition 5.9 (Chain-ladder projection as a random variable). $\hat{C}_{i,d+m} = C_{i,d} \prod_{k=d}^{d+m-1} \hat{f}_k$ with $d = n - 1 - i$.

Lemma 5.10. *$\hat{C}_{i,d+m}$ is $\mathcal{D}_{d+m}$-measurable.*

Proof. Induction on m; each new factor $\hat{f}_{d+m}$ is $\mathcal{D}_{d+m+1}$-measurable. □

Lemma 5.11 (Iterating (M1)). *For $d \le k$, $E[C_{i,k+1} \mid \mathcal{D}_d] = f_k \, E[C_{i,k} \mid \mathcal{D}_d]$, hence $E[C_{i,d+m} \mid \mathcal{D}_d] = C_{i,d} \prod_{k=d}^{d+m-1} f_k$.*

Proof. Tower property through $\mathcal{D}_k$, then (M1); induction on m. □

Theorem 5.12 (Mack 1993, Theorem 1). *Under (M1), with the partial products integrable and the column sums a.s. nonzero, $E[\hat{C}_{i,d+m} \mid \mathcal{D}_d] = C_{i,d} \prod_{k=d}^{d+m-1} f_k$, and therefore $E[\hat{C}_{i,n-1} \mid \mathcal{D}_d] = E[C_{i,n-1} \mid \mathcal{D}_d]$: the chain-ladder ultimate is a conditionally unbiased estimator of the true ultimate claims.*

Proof. Condition on $\mathcal{D}_{d+m}$: the partial product comes out and the last factor has conditional expectation f_{d+m} (Theorem 5.5); tower down to $\mathcal{D}_d$ and induct. Compare with Lemma 5.11. □

5.4 The variance assumptions and the estimation variance of $\hat{f}_k$

Definition 5.13 (Residuals, (M3) and (M2')). With $\varepsilon_{i,k} = C_{i,k+1} - f_k C_{i,k}$: (M3) $E[\varepsilon_{i,k}^2 \mid \mathcal{D}_k] = \sigma_k^2 C_{i,k}$, and (M2') $E[\varepsilon_{i,k}\varepsilon_{j,k} \mid \mathcal{D}_k] = 0$ for $i \neq j$, the conditional uncorrelatedness across accident years that independence of accident years supplies.

Lemma 5.14. *On $\{S_k \neq 0\}$, $\hat{f}_k - f_k = S_k^{-1} \sum_i \varepsilon_{i,k}$ and $(\hat{f}_k - f_k)^2 = S_k^{-2} \sum_i \sum_j \varepsilon_{i,k}\varepsilon_{j,k}$.*

Proof. $T_k - f_k S_k = \sum_i \varepsilon_{i,k}$; expand the square of the sum. □

Theorem 5.15 (Estimation variance of the development factor). *Under (M3) and (M2'), with the residual products integrable, on $\{S_k \neq 0\}$, $E[(\hat{f}_k - f_k)^2 \mid \mathcal{D}_k] = \sigma_k^2 / S_k$ (Mack 1993, proof of Theorem 3; Mack 1999, $\mathrm{s.e.}(\hat{f}_k)^2 = \hat{\sigma}_k^2 / S_k$).*

Proof. S_k^{-2} is $\mathcal{D}_k$-measurable and comes out; conditional expectation is linear over the double sum; cross terms vanish by (M2'); diagonal terms give $\sigma_k^2 \sum_i C_{i,k} = \sigma_k^2 S_k$. □

5.5 Unbiasedness of the variance estimator

Lemma 5.16 (Sum of squares around $\hat{f}_k$ via residuals)**.** *With all contributing $C_{i,k} \neq 0$ and $S_k \neq 0$, for any f, $\sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2 = \sum_i \varepsilon_i^2 / C_{i,k} - S_k(\hat{f}_k - f)^2$ where $\varepsilon_i = C_{i,k+1} - fC_{i,k}$.*

Proof. Two instances of Lemma 2.11 (centres $\hat{f}_k$ and f) and $T_k = \hat{f}_k S_k$. □

Definition 5.17. $\hat{\sigma}_k^2$ and the weighted sum of squares $W_k = \sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2$ as random variables; $\hat{\sigma}_k^2 = W_k/(n-k-2)$.

Theorem 5.18 (Unbiasedness of the variance estimator)**.** *Under (M3) and (M2'), with all contributing $C_{i,k}$ and S_k a.s. nonzero and the stated integrability, $E[W_k \mid \mathcal{D}_k] = (n-k-2)\,\sigma_k^2$ for $k+2 \leq n$, and hence $E[\hat{\sigma}_k^2 \mid \mathcal{D}_k] = \sigma_k^2$ for $k+3 \leq n$.*

Proof. By Lemma 5.16, $W_k = \sum_i \varepsilon_i^2/C_{i,k} - S_k(\hat{f}_k - f_k)^2$. Each $C_{i,k}^{-1}$ is $\mathcal{D}_k$-measurable, so (M3) gives σ_k^2 per term, $n-k-1$ terms in all; S_k is $\mathcal{D}_k$-measurable and Theorem 5.15 gives σ_k^2 for the subtracted term. □

5.6 Process variance and the exact MSEP (Theorem 3)

Lemma 5.19 (Conditional second moment)**.** *Under (M1) and (M3), $E[\varepsilon_{i,k} \mid \mathcal{D}_k] = 0$, $E[C_{i,k+1}^2 \mid \mathcal{D}_k] = f_k^2 C_{i,k}^2 + \sigma_k^2 C_{i,k}$, and for $d \leq k$, $E[C_{i,k+1}^2 \mid \mathcal{D}_d] = f_k^2 E[C_{i,k}^2 \mid \mathcal{D}_d] + \sigma_k^2 E[C_{i,k} \mid \mathcal{D}_d]$.*

Proof. Expand $C_{i,k+1} = f_k C_{i,k} + \varepsilon_{i,k}$; the cross term vanishes; tower property. □

Definition 5.20 (Process variance)**.** $V_0 = 0$, $V_{m+1} = f_{d+m}^2 V_m + \sigma_{d+m}^2\, c \prod_{l<m} f_{d+l}$; in closed form $V_m = \sum_{j<m} \big(\prod_{l=j+1}^{m-1} f_{d+l}\big)^2 \sigma_{d+j}^2\, c \prod_{l<j} f_{d+l}$.

Theorem 5.21 (Process variance along a row)**.** *Under (M1) and (M3), $E[C_{i,d+m}^2 \mid \mathcal{D}_d] - E[C_{i,d+m} \mid \mathcal{D}_d]^2 = V_m$ with $c = C_{i,d}$.*

Proof. Induction on m with Lemma 5.19 and Lemma 5.11. □

Lemma 5.22 (Conditional MSEP decomposition)**.** *For a $\mathcal{D}$-measurable predictor P of Y, $E[(P - Y)^2 \mid \mathcal{D}] = \big(E[Y^2 \mid \mathcal{D}] - E[Y \mid \mathcal{D}]^2\big) + (P - E[Y \mid \mathcal{D}])^2$.*

Proof. Expand the square; P comes out of the conditional expectation. □

Theorem 5.23 (Mack's Theorem 3, exact form)**.** *Let $\mathcal{D}$ be the observed data. Assume $\hat{C}_{i,d+m}$ is $\mathcal{D}$-measurable and that $\mathcal{D}$ adds nothing to $\mathcal{D}_d$ about row i's future (the conditional mean and second moment of $C_{i,d+m}$ given $\mathcal{D}$ agree with those given $\mathcal{D}_d$; this is what independence across accident years supplies). Then under (M1) and (M3)*

$$E\big[(\hat{C}_{i,d+m} - C_{i,d+m})^2 \mid \mathcal{D}\big] = V_m + \big(\hat{C}_{i,d+m} - C_{i,d} \textstyle\prod_{l<m} f_{d+l}\big)^2.$$

Proof. Lemma 5.22 with $P = \hat{C}_{i,d+m}$, then Theorem 5.21 and Lemma 5.11. □

Definition 5.24 (Mack's estimators of the two terms)**.** $\widehat{V}$ is V_m with $\hat{f}, \hat{\sigma}^2$ plugged in. The estimation-error term $\hat{C}_{i,n-1}^2 \sum_k \hat{\sigma}_k^2/(\hat{f}_k^2 S_k)$ is obtained from $(\hat{C} - M)^2$ by Mack's conditional-resampling step: replace $(\hat{f}_k - f_k)^2$ by σ_k^2/S_k (Theorem 5.15), drop the cross terms (Theorem 5.7), and plug in $\hat{f}, \hat{\sigma}^2$. This is a definition, not a theorem: it is the approximation in Mack (1993), and the exact conditional MSEP above is the object it approximates (Buchwalder, Bühlmann, Merz and Wüthrich 2006 estimate the same object differently).

Chapter 6

The variant catalogue: Mack versus conditional resampling

Write $a_k = \hat{\sigma}_k^2/(\hat{f}_k^2 S_k)$ for the relative estimation variance of development factor k along the row. Mack's estimation-error term is $\hat{C}_{i,n-1}^2 \sum_k a_k$; the conditional-resampling term of Buchwalder, Bühlmann, Merz and Wüthrich (2006), which Murphy (1994) also reaches, is $\hat{C}_{i,n-1}^2 \big(\prod_k (1 + a_k) - 1\big)$.

Definition 6.1. a_k and the conditional-resampling estimator $\hat{C}_{i,n-1}^2(\prod_k (1 + a_k) - 1)$.

Lemma 6.2 (Product minus sum)**.** *For $a_k \geq 0$: $1 + \sum a_k \leq \prod(1 + a_k) \leq \exp(\sum a_k)$; for two terms $(1 + a)(1 + b) - 1 - (a + b) = ab$; and $\prod(1 + a_k) - 1 - \sum a_k = 0$ when at most one a_k is nonzero.*

Proof. Induction over the index set for the Weierstrass inequality; $1 + x \leq e^x$ termwise for the exponential bound. □

Theorem 6.3 (Catalogue row: Mack is the first-order part of BBMW)**.** *With $a_k \geq 0$ along the row: (i) Mack $\leq$ BBMW; (ii) the difference is exactly $\hat{C}_{i,n-1}^2 \big(\prod_k (1 + a_k) - 1 - \sum_k a_k\big)$; (iii) the two coincide for the second-latest accident year, which has a single development factor; (iv) the difference is at most $\hat{C}_{i,n-1}^2 \big(e^{\sum a_k} - 1 - \sum a_k\big)$, second order in the relative estimation variances.*

Proof. Lemma 6.2 scaled by $\hat{C}_{i,n-1}^2$. □

Theorem 6.4 (Strict counterexample)**.** *There is a run-off triangle (four accident years, four development years, with unequal individual factors in the first two columns) and an accident year on which Mack's estimation-error term is strictly smaller than the conditional-resampling term: the two are different estimators, not two presentations of one. On the exhibited triangle $a_0 = 1/297$, $a_1 = 1/2809$, and the remainder is $\hat{C}^2 a_0 a_1 > 0$. Every number is checked by* `norm_num` *from the definitions.*

Proof. Evaluate the definitions on the triangle; apply Theorem 6.3(ii). □

6.1 The total reserve: Mack's Corollary

Definition 6.5 (Total-reserve estimators)**.** Mack's Corollary: $\widehat{\mathrm{msep}}(\hat{R}) = \sum_i \widehat{\mathrm{msep}}_i + \sum_i \hat{C}_i \big(\sum_{j>i} \hat{C}_j \big)\, 2 \sum_{k \in \mathrm{row}\ i} a_k$, the cross terms running over the older year's remaining factors. The conditional-resampling aggregate replaces $2 \sum_k a_k$ by $2(\prod_k (1 + a_k) - 1)$ (Buchwalder et al. 2006, Section 4.3).

Theorem 6.6 (Aggregated catalogue row)**.** *The aggregated estimation-error terms differ by* $\sum_i \big(\hat{C}_i^2 + 2\hat{C}_i \sum_{j>i} \hat{C}_j\big)\big(\prod_{k \in \mathrm{row}\ i}(1 + a_k) - 1 - \sum_{k \in \mathrm{row}\ i} a_k\big)$*, which is nonnegative when the ultimates and the* a_k *are: Mack's total is at most the resampling total, by the same row-wise second-order remainder as in the single-year case.*

Proof. Termwise algebra and Lemma 6.2. $\square$

Chapter 7

Non-vacuity: a nondegenerate model

Theorem 7.1 (A nondegenerate Mack model exists)**.** *There is a probability space, a random triangle with three accident years, and constants f_k, σ_k^2 with $\sigma_0^2 > 0$, satisfying (M1), (M3) and (M2'), in which the first development step is genuinely random. The model: eight equally likely outcomes, an independent ± 1 shock per accident year, $C_{i,0} = c$ and $C_{i,k} = cf^k(1 + s\,\xi_i)$ for $k \geq 1$, with $\mathcal{D}_0$ trivial and $\mathcal{D}_k$ everything for $k \geq 1$.*

Proof. Conditional expectations given the trivial σ-algebra are eight-term averages; the shocks sum to zero, have unit square, and are orthogonal. □

Theorem 7.2 (Headline theorems instantiated)**.** *On that model, $E[\hat{f}_0 \mid \mathcal{D}_0] = 2$ although $\hat{f}_0$ is random; $E[\hat{C}_{2,2} \mid \mathcal{D}_0] = E[C_{2,2} \mid \mathcal{D}_0]$; $E[(\hat{f}_0 - 2)^2 \mid \mathcal{D}_0] = \sigma_0^2/S_0$; and $E[\hat{\sigma}_0^2 \mid \mathcal{D}_0] = \sigma_0^2 = 4$.*

Proof. Direct instantiation of the general theorems; every hypothesis is discharged. □

Chapter 8

Reference computation

Mack's 1993 example (the Taylor and Ashe 1983 triangle) is reproduced to every printed digit (nine reserves, the total 18,681 thousand, all standard-error percentages) by a dependency-free script implementing the definitions above in floating point. The Lean definitions are over $\mathbb{R}$ and noncomputable; the script is the reference computation, not the Lean code.

Bibliography

[1] T. Mack, Distribution-free calculation of the standard error of chain ladder reserve estimates, ASTIN Bulletin 23 (1993) 213-225.

[2] T. Mack, The standard error of chain ladder reserve estimates: recursive calculation and inclusion of a tail factor, ASTIN Bulletin 29 (1999) 361-366.